EMCH 792: Optimal State Estimation
Homework 1
Written Portion Solutions

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Table of Contents

Exercise 1.2: Commuting Non-Diagonal Matrices (10 pts)

Find two 2×2 matrices A and B such that $A \neq B$, neither A nor B are diagonal, $A \neq cB$ for any scalar c, and $AB = BA$. Find the eigenvectors of A and B.

Step

Construction
We need two non-diagonal, non-scalar-multiple matrices that commute. A standard approach: choose matrices that share the same eigenvectors but have different eigenvalues.
Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

Verification of conditions:

- $A \neq B$ since $a_{12} = 1 \neq 2 = b_{12}$. ✓
- Neither is diagonal. ✓
- $A \neq cB$: if $A = cB$ then $c \cdot 1 = 1$ gives $c = 1$, but $c \cdot 2 = 1$ gives $c = 1/2$. Contradiction. ✓

Step

Checking $AB = BA$

$$AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

$AB = BA$. ✓

Step

Eigenvectors
Both A and B are upper triangular with repeated eigenvalue $\lambda = 1$.
For A: $(A - I)v = 0$ gives $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v = 0$, so $v_2 = 0$. Eigenvector: $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
For B: $(B - I)v = 0$ gives $\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} v = 0$, so $v_2 = 0$. Eigenvector: $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Results

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad AB = BA = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

Both A and B share the eigenvector $\boxed{\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}}$ (with eigenvalue $\lambda = 1$).

Exercise 1.3: Prove the Three Identities of Equation (1.26) (10 pts)

Prove:

1. $(AB)^T = B^T A^T$
2. $(AB)^{-1} = B^{-1} A^{-1}$
3. $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$

Step

Proof of $(AB)^T = B^T A^T$
Let A be $m \times n$ and B be $n \times p$. Then AB is $m \times p$.
The (i,j) entry of $(AB)^T$ is the (j,i) entry of AB:

$$[(AB)^T]_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk}\, b_{ki}$$

The (i,j) entry of $B^T A^T$:

$$[B^T A^T]_{ij} = \sum_{k=1}^{n} (B^T)_{ik}\, (A^T)_{kj} = \sum_{k=1}^{n} b_{ki}\, a_{jk} = \sum_{k=1}^{n} a_{jk}\, b_{ki}$$

These are identical, so $(AB)^T = B^T A^T$. ■

Step

Proof of $(AB)^{-1} = B^{-1} A^{-1}$
Assume A and B are square and invertible. We verify that $B^{-1}A^{-1}$ is the inverse of AB by showing it satisfies both conditions:
Right inverse:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Left inverse:

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Since the inverse is unique, $(AB)^{-1} = B^{-1}A^{-1}$. ■

Step

Proof of $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$
Let A be $m \times n$ and B be $n \times m$ (so both AB and BA are square).

$$\mathrm{Tr}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{k=1}^{n} a_{ik}\, b_{ki}$$

$$\mathrm{Tr}(BA) = \sum_{k=1}^{n} (BA)_{kk} = \sum_{k=1}^{n} \sum_{i=1}^{m} b_{ki}\, a_{ik}$$

Both double sums range over the same indices and contain the same terms $a_{ik}\, b_{ki}$, just with the order of summation swapped. Therefore $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. $\blacksquare$

Exercise 1.4: Partial Derivative of $\mathrm{Tr}(AB)$ with Respect to A (10 pts)

Find $\dfrac{\partial}{\partial A}\,\mathrm{Tr}(AB)$.

Step

Derivation
Let A be $m \times n$ and B be $n \times m$. Then:

$$\mathrm{Tr}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{k=1}^{n} a_{ik}\, b_{ki}$$

The matrix derivative $\dfrac{\partial}{\partial A}\,\mathrm{Tr}(AB)$ is defined as the $m \times n$ matrix whose (i, j) entry is:

$$\left[\frac{\partial}{\partial A}\,\mathrm{Tr}(AB)\right]_{ij} = \frac{\partial}{\partial a_{ij}}\,\mathrm{Tr}(AB)$$

Since $\mathrm{Tr}(AB) = \sum_{p=1}^{m} \sum_{k=1}^{n} a_{pk}\, b_{kp}$, the only term involving a_{ij} is $a_{ij}\, b_{ji}$. Therefore:

$$\frac{\partial}{\partial a_{ij}}\,\mathrm{Tr}(AB) = b_{ji}$$

This means $\left[\dfrac{\partial}{\partial A}\,\mathrm{Tr}(AB)\right]_{ij} = b_{ji} = (B^T)_{ij}$.

Results

$$\boxed{\frac{\partial}{\partial A}\,\mathrm{Tr}(AB) = B^T}$$

Exercise 1.6: Block Matrix Inverse (10 pts)

Suppose that A is invertible and

$$\begin{bmatrix} A & A \\ B & A \end{bmatrix} \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

Find B and C in terms of A.

Step

Expanding the Block Multiplication
Performing the block matrix–vector multiplication:

$$\begin{bmatrix} A \cdot A + A \cdot C \\ B \cdot A + A \cdot C \end{bmatrix} = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

This gives two equations:

$$A^2 + AC = 0 \tag{i}$$
$$BA + AC = I \tag{ii}$$

Step

Solving for C
From equation (i):

$$A^2 + AC = 0 \implies A(A + C) = 0$$

Since A is invertible, we can left-multiply by A^{-1}:

$$A + C = 0 \implies \boxed{C = -A}$$

Step

Solving for B
Substitute $C = -A$ into equation (ii):

$$BA + A(-A) = I \implies BA - A^2 = I$$

$$BA = I + A^2$$

Right-multiply by A^{-1}:

$$B = (I + A^2)A^{-1} = A^{-1} + A$$

Results

$$\boxed{C = -A}, \qquad \boxed{B = A^{-1} + A}$$

Verification: Substituting back into (i): $A^2 + A(-A) = A^2 - A^2 = 0$ ✓
Substituting back into (ii): $(A^{-1} + A)A + A(-A) = I + A^2 - A^2 = I$ ✓

Exercise 1.7: AB May Not Be Symmetric (10 pts)

Show that AB may not be symmetric even though both A and B are symmetric.

Step

Counterexample
Let
$$A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$
Both are symmetric: $A^T = A$ and $B^T = B$.
$$AB = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 2 & 6 \end{pmatrix}$$
Check symmetry:
$$(AB)^T = \begin{pmatrix} 1 & 2 \\ 4 & 6 \end{pmatrix} \neq \begin{pmatrix} 1 & 4 \\ 2 & 6 \end{pmatrix} = AB$$
Since $(AB)^T \neq AB$, the product AB is not symmetric. ■

Step

General Explanation
For AB to be symmetric, we would need $(AB)^T = AB$, i.e., $B^T A^T = AB$. When A and B are symmetric this becomes $BA = AB$. So AB is symmetric if and only if A and B commute. In general, symmetric matrices do not commute, so their product need not be symmetric.